

Original Lab Experiment - Full Code

Complete Python Code

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

iris = load_iris()
X = iris.data
y = iris.target

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, random_state=42
)

svm_model = SVC(kernel="linear")
svm_model.fit(X_train, y_train)

y_pred_svm = svm_model.predict(X_test)

print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
print("SVM Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_svm))

rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)

print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Random Forest Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_rf))

plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.heatmap(confusion_matrix(y_test,y_pred_svm),annot=True,cmap="Blues",fmt="d")
plt.title("SVM Confusion Matrix")

plt.subplot(1,2,2)
sns.heatmap(confusion_matrix(y_test,y_pred_rf),annot=True,cmap="Greens",fmt="d")
plt.title("Random Forest Confusion Matrix")

plt.tight_layout()
plt.show()
```

Expected Output

SVM Accuracy: 1.00

Random Forest Accuracy: 1.00

Both confusion matrices classify all samples correctly.

Output Graph

